
How to Create an Amazon EKS Cluster and Install kubectl

Step 1: Prerequisites

- AWS CLI installed and configured with appropriate permissions.
 - [eksctl](#) CLI tool installed (simplifies EKS cluster creation).
 - [kubectl](#) CLI installed (Kubernetes command-line tool).
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Step 2: Install kubectl

For Linux/macOS:

```
curl -LO "https://dl.k8s.io/release/$(curl -L -s https://dl.k8s.io/release/stable.txt)/bin/$(uname | tr '[:upper:]' '[:lower:]')/amd64/kubectl"
chmod +x kubectl
sudo mv kubectl /usr/local/bin/
kubectl version --client
```

For Windows (PowerShell):

```
curl.exe -LO "https://dl.k8s.io/release/v1.27.3/bin/windows/amd64/kubectl.exe"
```

Then add the downloaded [kubectl.exe](#) to your system PATH.

Step 3: Install eksctl

eksctl is the easiest way to create an EKS cluster.

For Linux/macOS:

```
curl --silent --location  
"https://github.com/weaveworks/eksctl/releases/latest/download/eksctl_$(uname)_amd64.tar.g  
z" | tar xz -C /tmp  
sudo mv /tmp/eksctl /usr/local/bin  
eksctl version
```

For Windows:

Download the **eksctl.exe** from the [eksctl releases page](#) and add it to your PATH.

Step 4: Create an EKS Cluster with eksctl

Run this command to create a simple EKS cluster:

```
eksctl create cluster --name my-eks-cluster --region us-west-2 --nodes 2 --nodes-min 1  
--nodes-max 3 --managed
```

Parameters:

- **--name**: Name of your cluster.
- **--region**: AWS region (choose your preferred).
- **--nodes**: Number of worker nodes.
- **--nodes-min** and **--nodes-max**: Auto scaling limits.
- **--managed**: Use managed node groups.

This will take around 10-15 minutes.

Step 5: Configure kubectl to Use Your Cluster

`eksctl` automatically updates your kubeconfig file to access the cluster.

You can verify with:

```
kubectl get nodes
```

This should list your worker nodes in the cluster.

Optional: Manually update kubeconfig (if needed)

```
aws eks --region us-west-2 update-kubeconfig --name my-eks-cluster
```

Summary

Step	Command/Action
Install kubectl	Download from Kubernetes releases and add to PATH
Install eksctl	Download from GitHub and add to PATH
Create cluster	<code>eksctl create cluster --name my-eks-cluster ...</code>
Verify kubectl access	<code>kubectl get nodes</code>
